



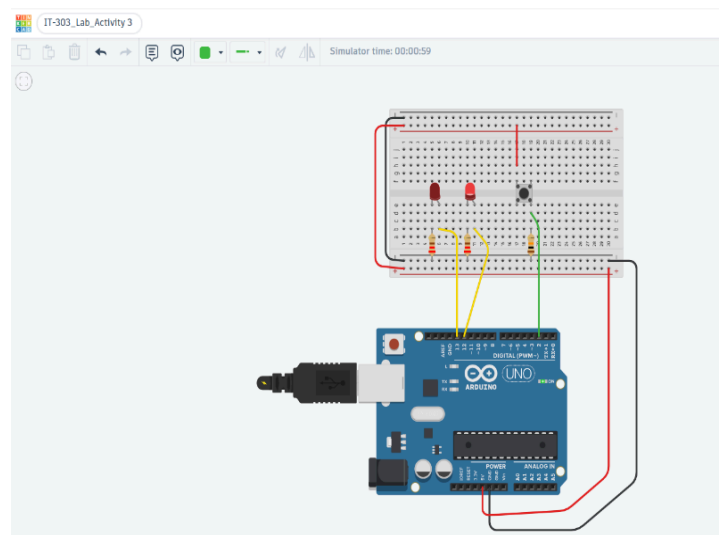
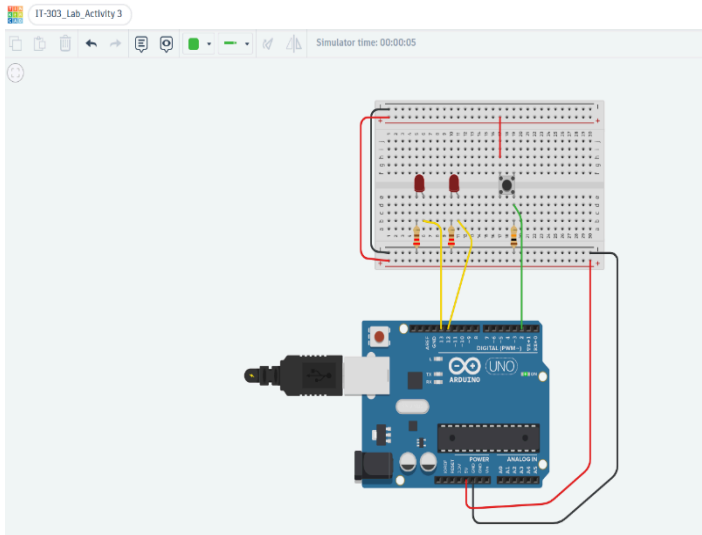
Activity 4: Two architectures, comparison table, justified choice

I. Requirement Statement.

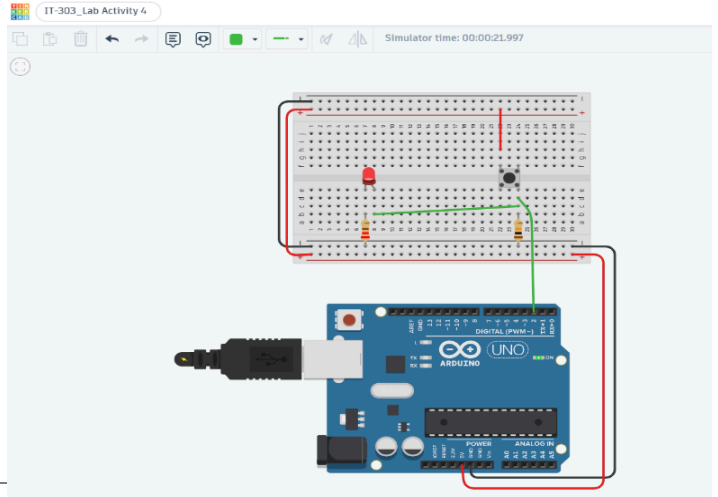
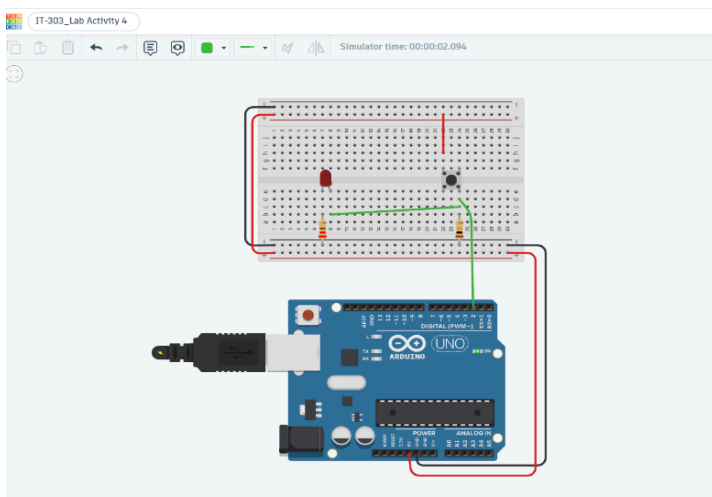
The LED will turn ON when pressing or hold the button, and turn off every release. The system records it each change.

II. Screenshots of both systems working.

SYSTEM 1:



SYSTEM 2:





III. The completed comparison table, all four criteria.

Criteria	System 1 With Arduino controlled	System 2 Direct to button controlled
Parts Count	Arduino, pushbutton, 2 LEDs, 3 resistors, connecting wires	Arduino, pushbutton, 1 LED, 2 resistors, connecting wires
Junctions & Connections	Button to Arduino input, LEDs to Arduino output pins	Button to LED directly; one wire from button only to Arduino input pin
Cost of Change: Make indicator blink	Very easy - update the code only, add delay() and toggle logic. No hardware changes needed.	Difficult - cannot be done through code alone. Requires rewiring and adding components (transistor/relay) since Arduino does not control the LED.
Failure Behaviour: Controller stopped	Entire system fails - LED stays off; button has no effect at all.	LED still works through button turns LED on/off normally and only recording to Serial Monitor stops.

IV. Your two-sentence justified choice.

If I had to choose between the two systems, I would choose **System 1** because the requirement states that the system must record every change, and System 1 guarantees that every action is both triggered and logged by the controller. Then **System 2** is not a choice because changes can happen without being recorded whenever the controller is not running.



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